Project: Worked examples
Title: Thick-walled cylinder (Lamé)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Radial, hoop and axial stresses at the bore and the outer surface of a thick-walled cylinder under internal and external pressure, the von Mises and Tresca equivalents, utilisation against an allowable, and the thin-wall estimate for comparison.

WHAT IT CALCULATES	RELATION
Lamé constants from the bore and outside pressures	$A = (a^2 p_i - b^2 p_o)/(b^2 - a^2)$; $B = a^2 b^2 (p_i - p_o)/(b^2 - a^2)$
Radial and hoop stress at radius r	$\sigma_r = A - B/r^2$; $\sigma_\theta = A + B/r^2$
Axial stress, closed or open ends	$\sigma_z = A$ (closed); $\sigma_z = 0$ (open)
Equivalent stresses from the three principal stresses	von Mises; Tresca = the largest principal difference

Basis: Thick-walled cylinders[1](Ch. 13, Table 13.5); Stresses in pressurised cylinders[2](§3-14).

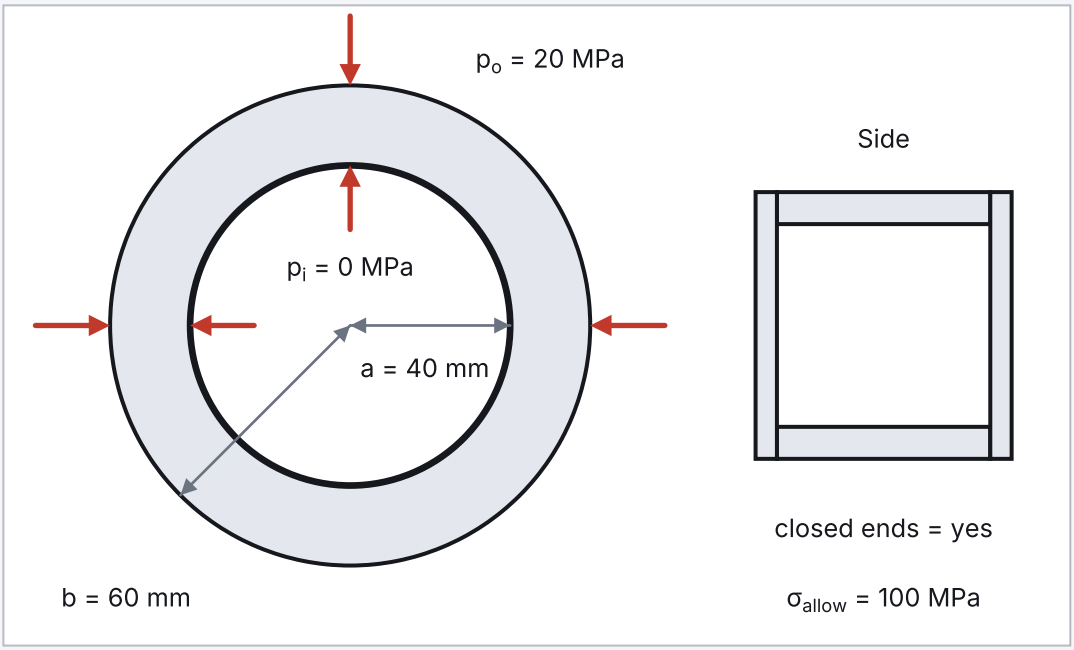


Figure 1 — A thick-walled cylinder in section, bore radius a, outer radius b, pressure inside and out; closed ends carry the pressures axially. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
a	Inner radius	40	mm	drawing: DRG-CYL-02 rev B	0.1 … 5000 mm
b	Outer radius	60	mm	drawing: DRG-CYL-02 rev B	0.1 … 5000 mm
p_i	Internal pressure	0	MPa	assumption: Vented bore.	0 … 1000 MPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
p_o	External pressure	20	MPa	spec: SPEC-SUB-1 §2 external pressure	0 … 1000 MPa
closed	Closed ends (axial stress carried by the wall)	yes		drawing: DRG-CYL-02 rev B	any
E	Young's modulus (the elastic collapse pressure under external pressure only)	207	GPa	default	1 … 500 GPa
v	Poisson's ratio (the elastic collapse pressure under external pressure only)	0.3		default	0.1 … 0.49
σ_{allow}	Allowable stress	100	MPa	spec: SPEC-SUB-1 §4	1 … 3000 MPa

3. Assumptions

A1. Long cylinder, elastic, isotropic; stresses vary with radius only. (Lamé solution; end effects neglected.)
A2. Closed ends carry the pressure end load as a uniform axial stress. (σ_z = A.)
A3. p_i (Internal pressure) is taken as 0 MPa. (Vented bore.)

4. Working

[9] A = (a^2 * p_i - b^2 * p_o) / (b^2 - a^2) [1](Table 13.5)
= ((40 mm)^2 × 0 MPa − (60 mm)^2 × 20 MPa) / ((60 mm)^2 − (40 mm)^2)
= -36 MPa
Lamé constant A

[10] B = a^2 * b^2 * (p_i - p_o) / (b^2 - a^2) [1](Table 13.5)
= (40 mm)^2 × (60 mm)^2 × (0 MPa − 20 MPa) / ((60 mm)^2 − (40 mm)^2)
= -57600 MPa·mm²
Lamé constant B

[11] $\sigma_{r,a}$ = A - B / a^2
= (-36 MPa) − (-57600 MPa·mm²) / (40 mm)^2
= 0 MPa
Radial stress at the bore

[12] $\sigma_{t,a}$ = A + B / a^2
= -36 MPa + (-57600 MPa·mm²) / (40 mm)^2
= -72 MPa
Hoop stress at the bore

[13] $\sigma_{r,b}$ = A - B / b^2
= (-36 MPa) − (-57600 MPa·mm²) / (60 mm)^2
= -20 MPa
Radial stress at the outer surface

[14] $\sigma_{t,b}$ = A + B / b^2
= -36 MPa + (-57600 MPa·mm²) / (60 mm)^2
= -52 MPa
Hoop stress at the outer surface

[15] σ_z = A
= -36 MPa
= -36 MPa
Axial stress (closed ends)

[16] $\sigma_{vm,a}$ = sqrt(((($\sigma_{t,a}$ - $\sigma_{r,a}$)^2 + ($\sigma_{r,a}$ - σ_z)^2 + (σ_z - $\sigma_{t,a}$)^2) / 2)
= sqrt((((-72 MPa) − 0 MPa)^2 + (0 MPa − (-36 MPa))^2 + ((-36 MPa) − (-72 MPa))^2) / 2)

= 62.354 MPa

von Mises stress at the bore

[17] $\sigma_{vm,b} = \text{sqrt}(((\sigma_{t,b} - \sigma_{r,b})^2 + (\sigma_{r,b} - \sigma_z)^2 + (\sigma_z - \sigma_{t,b})^2) / 2)$

= sqrt((((-52 MPa) - (-20 MPa))^2 + ((-20 MPa) - (-36 MPa))^2 + ((-36 MPa) - (-52 MPa))^2) /

= 27.713 MPa

von Mises stress at the outer surface

[18] $\sigma_{tr,a} = \max(\sigma_{t,a}, \sigma_{r,a}, \sigma_z) - \min(\sigma_{t,a}, \sigma_{r,a}, \sigma_z)$

= max(-72 MPa, 0 MPa, -36 MPa) - min(-72 MPa, 0 MPa, -36 MPa)

= 72 MPa

Tresca stress at the bore

[19] $\sigma_{tr,b} = \max(\sigma_{t,b}, \sigma_{r,b}, \sigma_z) - \min(\sigma_{t,b}, \sigma_{r,b}, \sigma_z)$

= max(-52 MPa, -20 MPa, -36 MPa) - min(-52 MPa, -20 MPa, -36 MPa)

= 32 MPa

Tresca stress at the outer surface

[20] $\sigma_{vm,max} = \max(\sigma_{vm,a}, \sigma_{vm,b})$

= max(62.354 MPa, 27.713 MPa)

= 62.354 MPa

Maximum von Mises stress

[21] $u = \sigma_{vm,max} / \sigma_{allow}$

= 62.354 MPa / 100 MPa

= 0.62354

Utilisation

[22] $\sigma_{thin} = (p_i - p_o) * (a + b) / 2 / (b - a)$

= (0 MPa - 20 MPa) × (40 mm + 60 mm) / 2 / (60 mm - 40 mm)

= -50 MPa

Thin-wall hoop estimate at the mean radius

[23] $k_{wall} = b / a$

= 60 mm / 40 mm

= 1.5

Wall ratio b/a

5. Results

SYMBOL	VALUE	DESCRIPTION
$\sigma_{r,a}$	0 MPa	Radial stress, bore
$\sigma_{t,a}$	-72.0 MPa	Hoop stress, bore
$\sigma_{r,b}$	-20.0 MPa	Radial stress, outer
$\sigma_{t,b}$	-52.0 MPa	Hoop stress, outer
σ_z	-36.0 MPa	Axial stress
$\sigma_{vm,a}$	62.4 MPa	von Mises, bore
$\sigma_{vm,b}$	27.7 MPa	von Mises, outer
$\sigma_{tr,a}$	72.0 MPa	Tresca, bore
$\sigma_{tr,b}$	32.0 MPa	Tresca, outer
$\sigma_{vm,max}$	62.4 MPa	Maximum von Mises
σ_{thin}	-50.0 MPa	Thin-wall hoop estimate
k_{wall}	1.5	Wall ratio
u	0.62	Utilisation (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	von Mises against allowable	$u = 0.62$	$u \le 1$	0.62	

7. Applicability

None.

Defaults used (inputs left empty):
E: not entered, so the default of 207 GPa is used.
v: not entered, so the default of 0.3 is used.

8. Verification

`Template mech.thick-cylinder@2.1.1 Content hash aed5864571238a05 Engine calc-engine/1.0.0`
`Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0`

9. References

1. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.